Tutorial 09
Properties of Relations

Example 8.1.5 – Inverse of an infinite relation

Recap:

Definition

Let R be a relation from A to B. Define the inverse relation R^{-1} from B to A as follows:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in R\}.$$

Problem:

Define a relation R from $\mathbf{R}$ to $\mathbf{R}$ as follows: For every ordered pair $(x, y) \in \mathbf{R} \times \mathbf{R}$,

$$x\, R\, y \quad \Leftrightarrow \quad y = 2|x|.$$

Draw the graphs of R and R^{-1} in the Cartesian plane. Is R^{-1} a function?

Example 8.1.5 – Inverse of an infinite relation

Solution A point (v, u) is on the graph of R^{-1} if, and only if, (u, v) is on the graph of R. Note that if $x \geq 0$, then the graph of $y = 2|x| = 2x$ is a straight line with slope 2. And if $x < 0$, then the graph of $y = 2|x| = 2(-x) = -2x$ is a straight line with slope -2. Some sample values are tabulated and the graphs are shown below.

$$R = \{(x, y) \mid y = 2|x|\}$$

x	y
0	0
1	2
-1	2
2	4
-2	4

1st coordinate 2nd coordinate

$$R^{-1} = \{(y, x) \mid y = 2|x|\}$$

y	x
0	0
2	1
2	-1
4	2
4	-2

1st coordinate 2nd coordinate

Example 8.1.5 – Inverse of an infinite relation

$R = \{(x, y) \mid y = 2|x|\}$

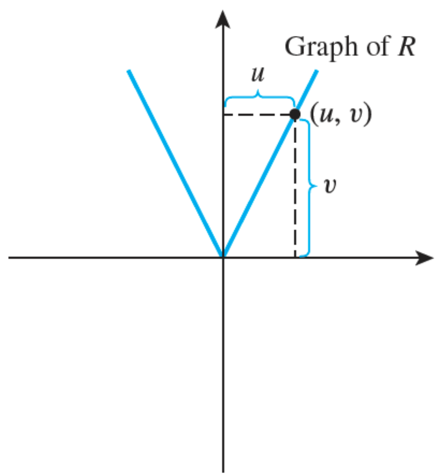
x	y
0	0
1	2
−1	2
2	4
−2	4

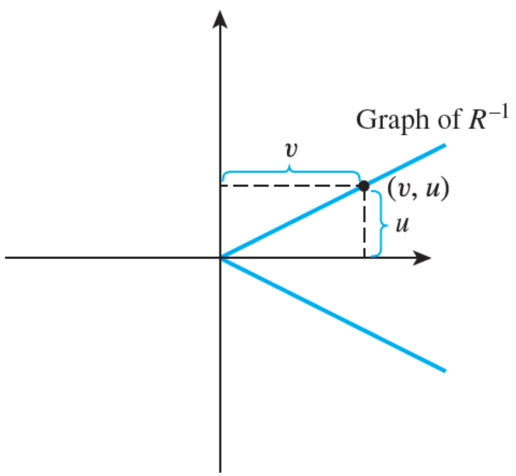
1st coordinate 2nd coordinate

$R^{-1} = \{(y, x) \mid y = 2|x|\}$

y	x
0	0
2	1
2	−1
4	2
4	−2

1st coordinate 2nd coordinate





R^{-1} is not a function because, for instance, both $(2, 1)$ and $(2, -1)$ are in R^{-1}. ■

Example 8.2.4 – Properties of Congruence Modulo 3

Recap:

Definition

Let R be a relation on a set A.

1. R is **reflexive** if, and only if, for every $x \in A$, $x \, R \, x$.
2. R is **symmetric** if, and only if, for every $x, y \in A$, if $x \, R \, y$ then $y \, R \, x$.
3. R is **transitive** if, and only if, for every $x, y, z \in A$, if $x \, R \, y$ and $y \, R \, z$ then $x \, R \, z$.

Problem:

Define a relation T on $\mathbf{Z}$ (the set of all integers) as follows: For all integers m and n,

$$\boxed{m \, T \, n \quad \Leftrightarrow \quad 3 \mid (m - n).}$$

This relation is called **congruence modulo 3**.

a. Is T reflexive?

b. Is T symmetric?

c. Is T transitive?

Example 8.2.4 – Properties of Congruence Modulo 3

Solution

a. ***T is reflexive*:** To show that T is reflexive, it is necessary to show that

$$\text{For every } m \in \mathbf{Z}, \quad m\,T\,m.$$

By definition of T, this means that

$$\text{For every } m \in \mathbf{Z}, \quad 3 \mid (m - m),$$

which is true because $m - m = 0$ and $3 \mid 0$ (since $0 = 3 \cdot 0$). Hence T is reflexive.

Example 8.2.4 – Properties of Congruence Modulo 3

b. ***T is symmetric*:** To show that T is symmetric, it is necessary to show that

$$\text{For every } m, n \in \mathbf{Z}, \quad \textbf{if } m\ T\ n \text{ then } n\ T\ m.$$

By definition of T this means that

$$\text{For every } m, n \in \mathbf{Z}, \quad \textbf{if } 3 \mid (m-n) \text{ then } 3 \mid (n-m).$$

Is this true? Suppose m and n are particular but arbitrarily chosen integers such that $3 \mid (m-n)$. Must it follow that $3 \mid (n-m)$? *[In other words, can we find an integer so that $n-m = 3\cdot$(that integer)?]* By definition of "divides," since

$$3 \mid (m-n),$$

then

$$m - n = 3k \quad \text{for some integer } k.$$

The crucial observation is that $n-m = -(m-n)$. Hence, you can multiply both sides of this equation by -1 to obtain

$$-(m-n) = -3k,$$

which is equivalent to

$$n - m = 3(-k).$$

[Thus we have found an integer, $-k$, so that $n-m = 3\cdot$(that integer).]

Since $-k$ is an integer, this equation shows that

$$3 \mid (n-m).$$

It follows that T is symmetric.

Example 8.2.4 – Properties of Congruence Modulo 3

c. ***T is transitive*:** To show that T is transitive, it is necessary to show that

$$\text{For every } m, n, p \in \mathbf{Z}, \quad \textbf{if } m\ T\ n \text{ and } n\ T\ p \text{ then } m\ T\ p.$$

By definition of T this means that

$$\text{For every } m, n \in \mathbf{Z}, \quad \textbf{if } 3 \mid (m-n) \text{ and } 3 \mid (n-p) \text{ then } 3 \mid (m-p).$$

Is this true? Suppose m, n, and p are particular but arbitrarily chosen integers such that $3 \mid (m-n)$ and $3 \mid (n-p)$. Must it follow that $3 \mid (m-p)$? *[In other words, can we find an integer so that $m-p = 3\cdot$(that integer)?]* By definition of "divides," since

$$3 \mid (m-n) \quad \text{and} \quad 3 \mid (n-p),$$

then

$$m-n = 3r \quad \text{for some integer } r,$$

and

$$n-p = 3s \quad \text{for some integer } s.$$

Example 8.2.4 – Properties of Congruence Modulo 3

$$m - n = 3r \quad \text{for some integer } r,$$

and

$$n - p = 3s \quad \text{for some integer } s.$$

The crucial observation is that $(m - n) + (n - p) = m - p$. Add these two equations together to obtain

$$(m - n) + (n - p) = 3r + 3s,$$

which is equivalent to

$$m - p = 3(r + s).$$

[Thus we have found an integer so that $m - p = 3 \cdot$ (that integer).] Since r and s are integers, $r + s$ is an integer. So this equation shows that

$$3 \mid (m - p).$$

It follows that T is transitive.

Exercise – Equivalence & Equivalence Class

Recap:

Definition

Let A be a set and R a relation on A. R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Definition

Suppose A is a set and R is an equivalence relation on A. For each element a in A, the **equivalence class of *a***, denoted **[*a*]** and called the **class of *a*** for short, is the set of all elements x in A such that x is related to a by R.

In symbols:

$$[a] = \{x \in A \mid x\, R\, a\}$$

Exercise – Equivalence & Equivalence Class

Problem:

Let $A = \mathbf{Z}^+ \times \mathbf{Z}^+$. Define a relation R on A as follows: For every (a, b) and (c, d) in A,

$$(a, b)\, R\, (c, d) \quad \Leftrightarrow \quad a + d = c + b.$$

Q(1) Prove that the relation is an equivalence relation.
Q(2) Describe the distinct equivalence classes of the relation.

Exercise – Equivalence & Equivalence Class

Solution

(1) The goal is to prove relation E is reflexive, symmetric, and transitive.

(a) Prove that R is reflexive

A relation R is **reflexive** if every element is related to itself. That is:

$$\forall (a, b) \in A, \quad (a, b)\ R\ (a, b)$$

Let's check:

$$(a + b) = (a + b) \Rightarrow (a, b)\ R\ (a, b)$$

✅ So, the relation is reflexive.

Exercise – Equivalence & Equivalence Class

Solution

(b) Prove that R is symmetric

A relation R is **symmetric** if:

$$(a, b)\ R\ (c, d) \Rightarrow (c, d)\ R\ (a, b)$$

We're given:

$$a + d = c + b$$

This implies:

$$c + b = a + d \Rightarrow (c, d)\ R\ (a, b)$$

✅ So, the relation is symmetric.

Exercise – Equivalence & Equivalence Class

Solution

(c) Prove that R is transitive

A relation R is **transitive** if:

$$(a, b)\ R\ (c, d) \text{ and } (c, d)\ R\ (e, f) \Rightarrow (a, b)\ R\ (e, f)$$

From the definition:

- $(a, b)R(c, d) \Rightarrow a + d = c + b$
- $(c, d)R(e, f) \Rightarrow c + f = e + d$

We aim to prove:

$$(a, b)R(e, f) \Rightarrow a + f = e + b$$

From the two assumptions:

1. $a + d = c + b$ → rearranged: $a - c = b - d$
2. $c + f = e + d$ → rearranged: $c - e = d - f$

Add both equations:

$$(a - c) + (c - e) = (b - d) + (d - f) \Rightarrow a - e = b - f \Rightarrow a + f = e + b \Rightarrow (a, b)R(e, f)$$

✅ So, the relation is transitive.

Exercise – Equivalence & Equivalence Class

Solution

(2) Describe distinct equivalence classes of the relation.

From earlier simplification, the relation can be rewritten as:

$$(a, b)\ R\ (c, d) \iff a - b = c - d$$

Thus, the equivalence classes are defined by the difference $a - b$. For each integer $k \in \mathbb{Z}$, define:

$$C_k = \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid a - b = k\}$$

These are the **distinct equivalence classes** of R, one for each integer k, where elements share the same difference between their first and second components.

✅ So, the equivalence classes are all sets of pairs with the same difference $a - b$, one for each $k \in \mathbb{Z}$.

Application of Partial Order Relations

Note that the following defines a partial order relation on the set of courses required for a university degree:

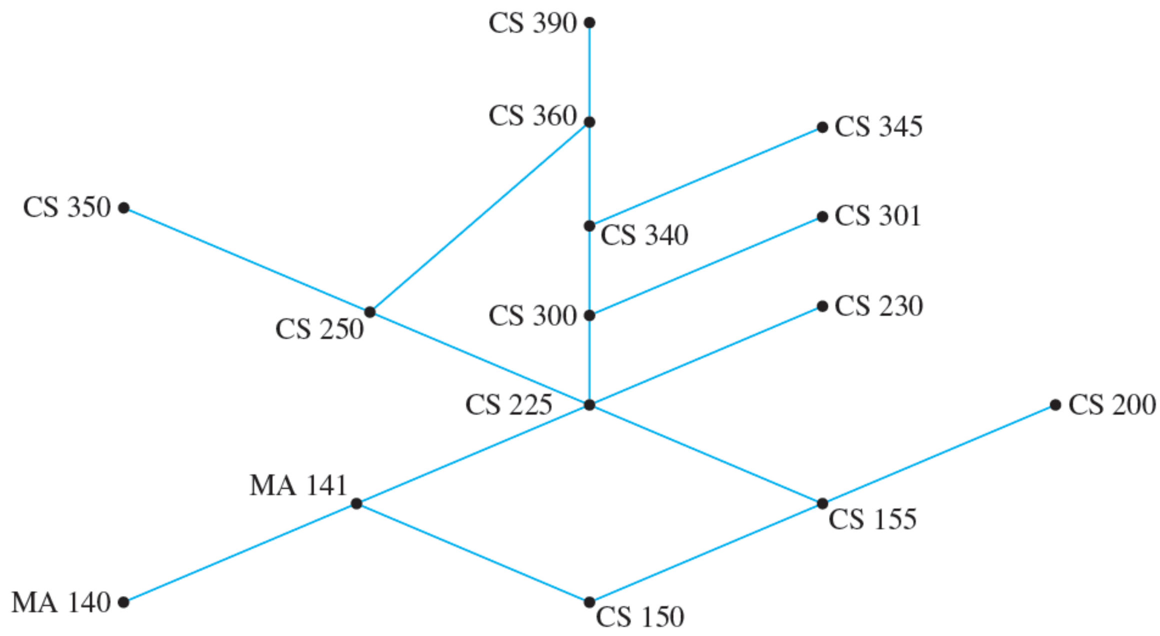
For all required courses *x* and *y*,

$$x \preceq y \quad \Leftrightarrow \quad x = y \quad \text{or} \quad x \text{ is a prerequisite for } y.$$

If the Hasse diagram for the relation is drawn, then many questions can be answered easily.

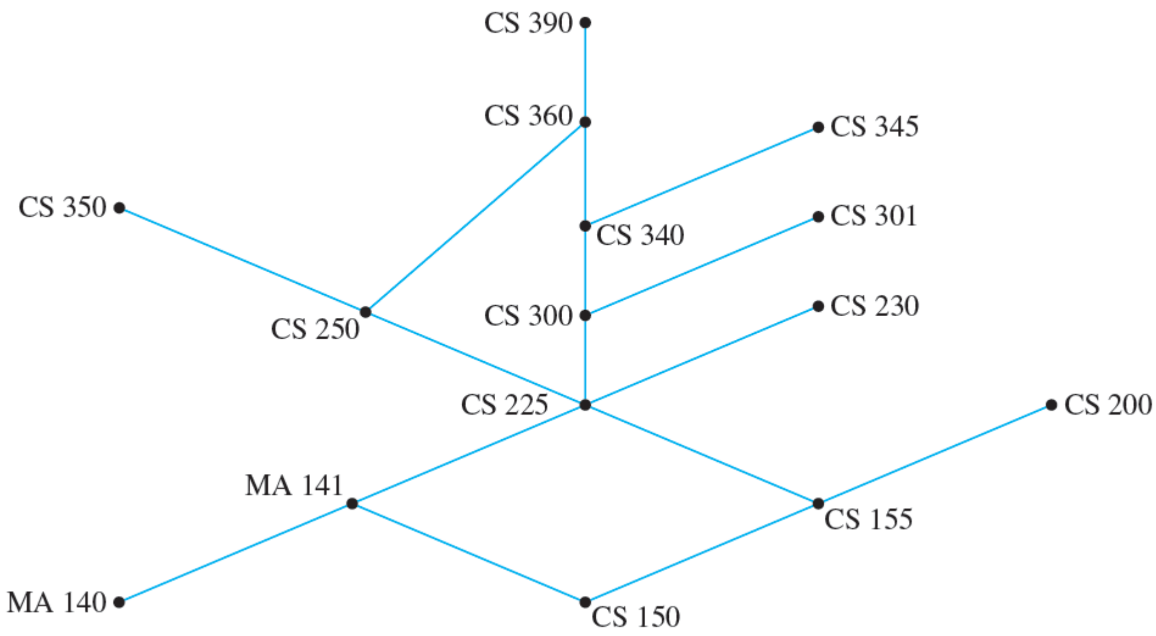
Application of Partial Order Relations

For instance, consider the Hasse diagram for the requirements at a particular university, which is shown in the below Figure.



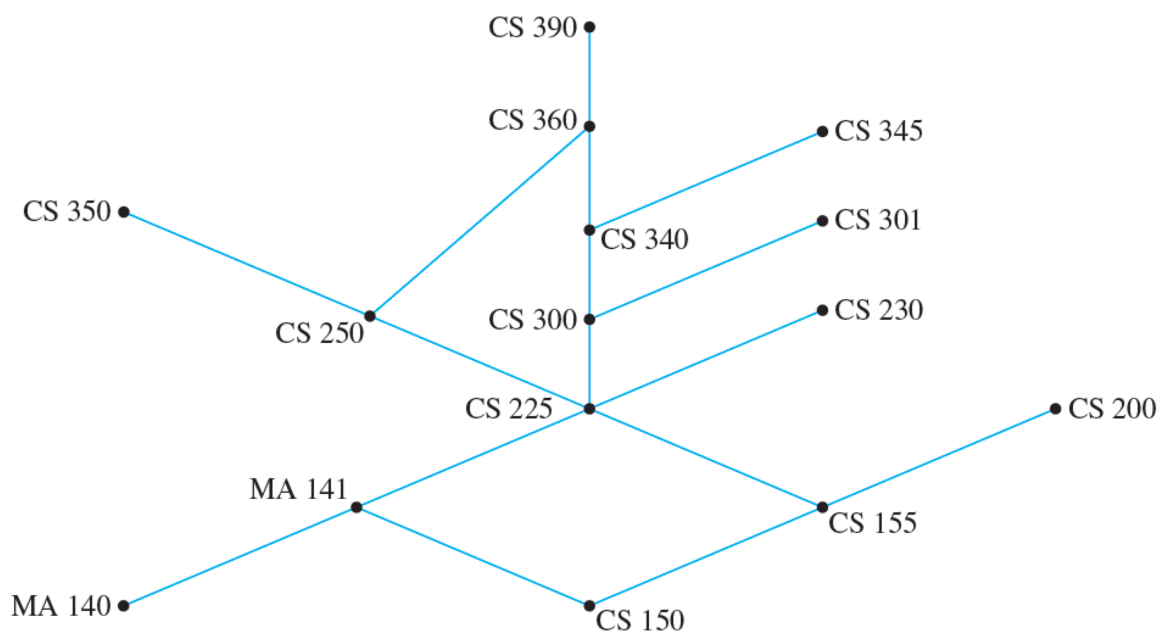
Application of Partial Order Relations

Q1. What is the minimum number of school terms needed to complete the degree requirements?



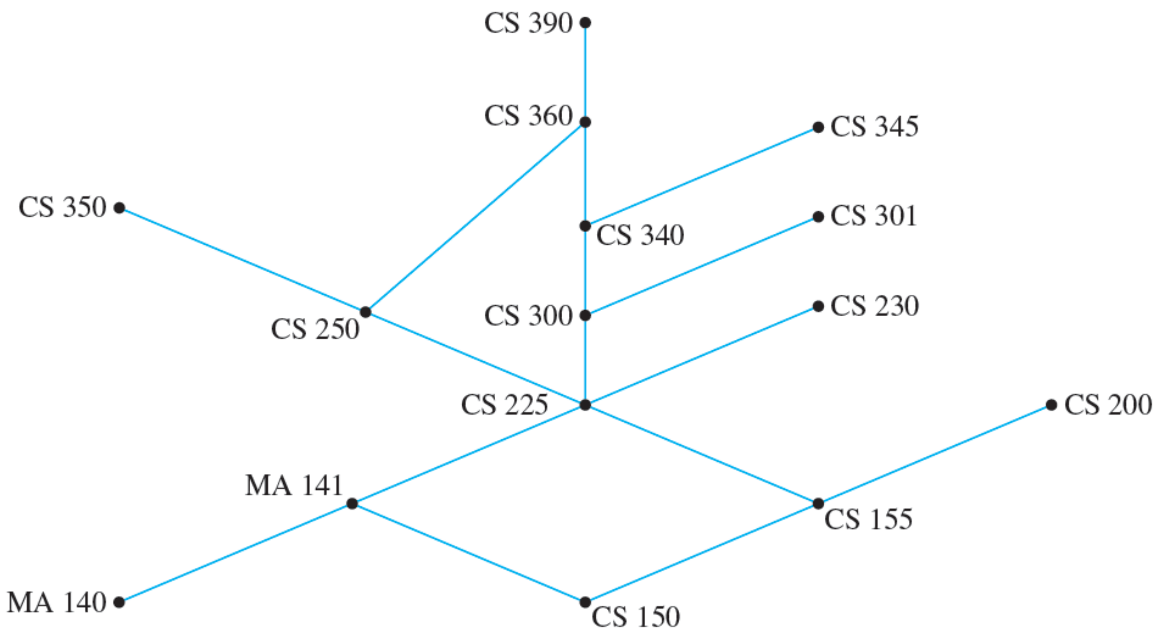
Application of Partial Order Relations

The minimum number of school terms needed to complete the requirements is the size of a longest chain, which is 7 (150, 155, 225, 300, 340, 360, 390, for example).



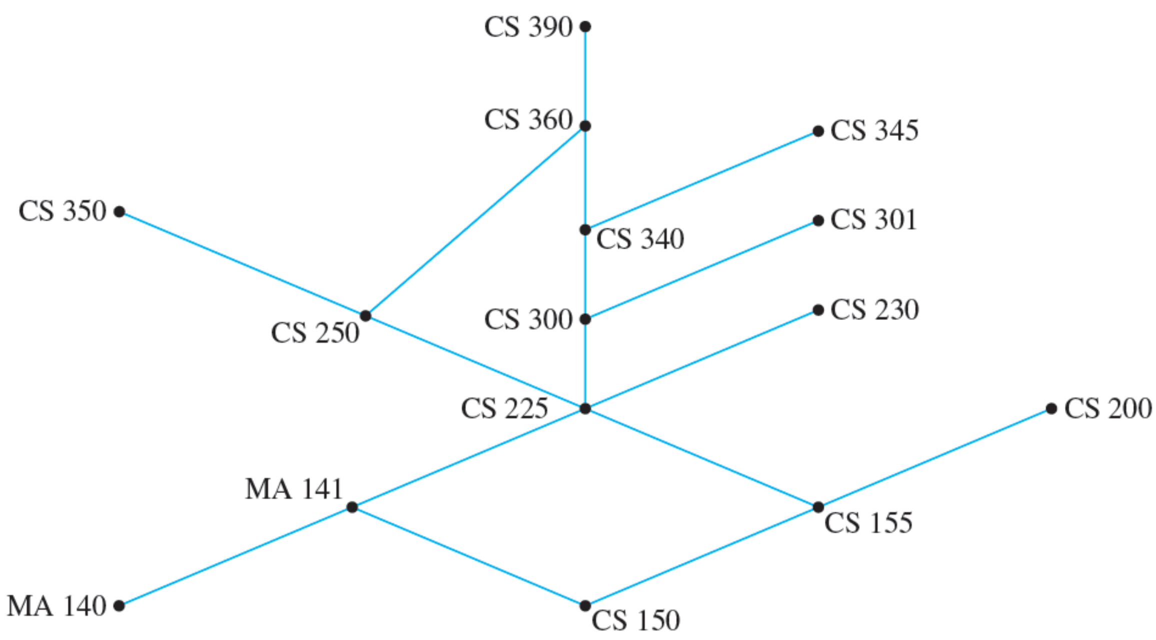
Application of Partial Order Relations

Q2. What is the maximum number of courses that could be taken in the same term?



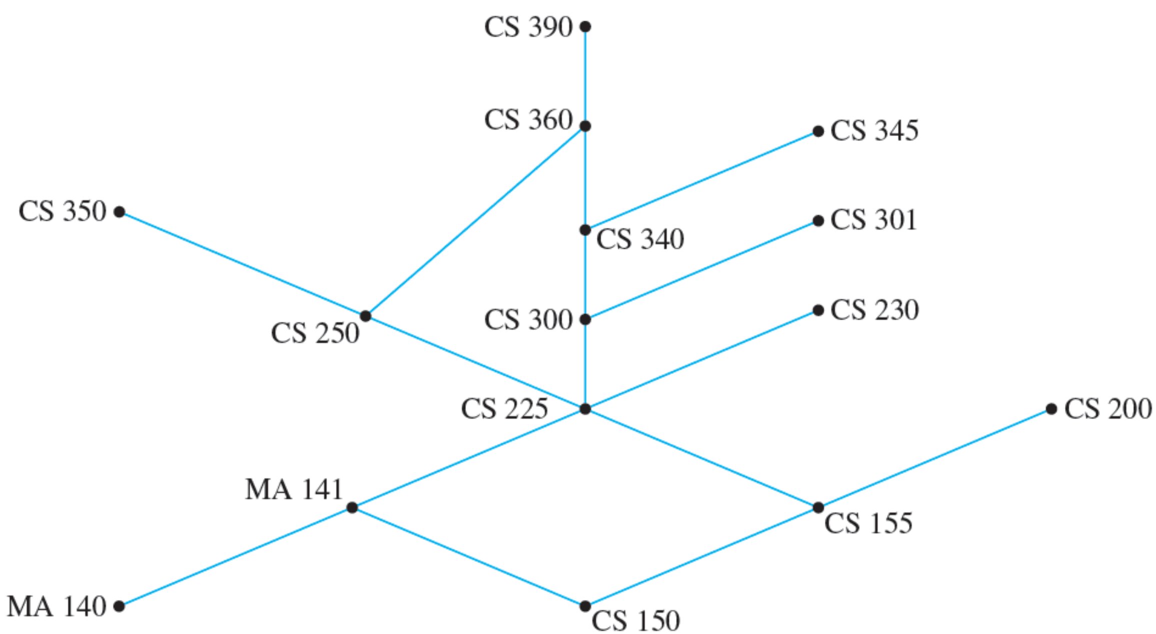
Application of Partial Order Relations

The maximum number of courses that could be taken in the same term (assuming the university allows it) is the maximum number of noncomparable courses, which is 6 (350, 360, 345, 301, 230, 200, for example).



Application of Partial Order Relations

A student could take the courses in a sequence determined by constructing a topological sorting for the set. (One such sorting is 140, 150, 141, 155, 200, 225, 230, 300, 250, 301, 340, 345, 350, 360, 390. There are many others.)



PERT and CPM

Two important and widely used applications of partial order relations are **PERT** (Program Evaluation and Review Technique) and **CPM** (Critical Path Method).

These techniques were developed in the 1950s as planners came to grips with the complexities of scheduling the individual activities needed to complete very large projects, and although they are very similar, their developments were independent.

PERT and CPM

PERT was developed by the U.S. Navy to help organize the construction of the Polaris submarine, and CPM was developed by the E. I. Du Pont de Nemours company for scheduling chemical plant maintenance.

Here is a somewhat simplified example of the way the techniques work.

Example 8.5.12 – *A Job Scheduling Problem*

At an automobile assembly plant, the job of assembling an automobile can be broken down into these tasks:

1. Build frame.
2. Install engine, power train components, gas tank.
3. Install brakes, wheels, tires.
4. Install dashboard, floor, seats.
5. Install electrical lines.
6. Install gas lines.

Example 8.5.12 – *A Job Scheduling Problem* continued

7. Install brake lines.
8. Attach body panels to frame.
9. Paint body.

Certain of these tasks can be carried out at the same time, whereas some cannot be started until other tasks are finished.

Example 8.5.12 – *A Job Scheduling Problem* continued

Table 8.5.1 summarizes the order in which tasks can be performed and the time required to perform each task.

Task	Immediately Preceding Tasks	Time Needed to Perform Task
1		7 hours
2	1	6 hours
3	1	3 hours
4	2	6 hours
5	2, 3	3 hours
6	4	1 hour
7	2, 3	1 hour
8	4, 5	2 hours
9	6, 7, 8	5 hours

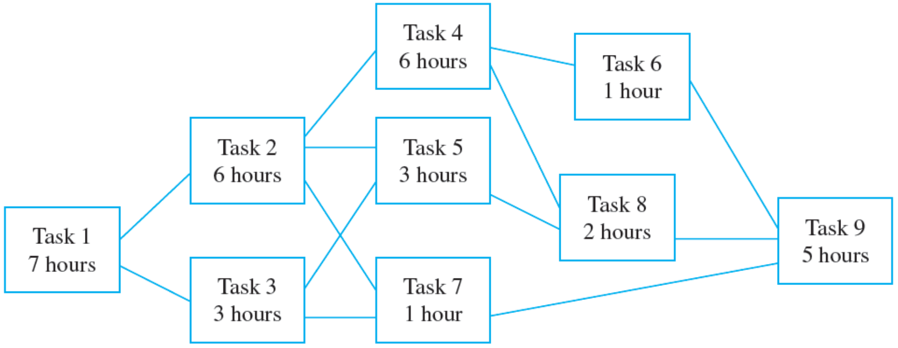
Table 8.5.1

Example 8.5.12 – *A Job Scheduling Problem* continued

Let T be the set of all tasks, and consider the partial order relation $\preceq$ defined on T as follows: For all tasks x and y in T,

$$\boxed{x \preceq y \quad \Leftrightarrow \quad x = y \text{ or } x \text{ precedes } y.}$$

If the Hasse diagram of this relation is turned sideways (as is customary in PERT and CPM analysis), it has the appearance shown on the right side.



Example 8.5.12 – *A Job Scheduling Problem* continued

What is the minimum time required to assemble a product? You can determine this by working from left to right across the diagram, noting for each task (say, just above the box representing that task) the minimum time needed to complete that task starting from the beginning of the assembly process.

For instance, you can put a 7 above the box for task 1 because task 1 requires 7 hours.

Example 8.5.12 – *A Job Scheduling Problem* continued

Task 2 requires completion of task 1 (7 hours) plus 6 hours for itself, so the minimum time required to complete task 2, starting at the beginning of the assembly process, is
$7 + 6 = 13$ hours.

So you can put a 13 above the box for task 2. Similarly, you can put a 10 above the box for task 3 because $7 + 3 = 10$.

Now consider what number you should write above the box for task 5.

Example 8.5.12 – *A Job Scheduling Problem* continued

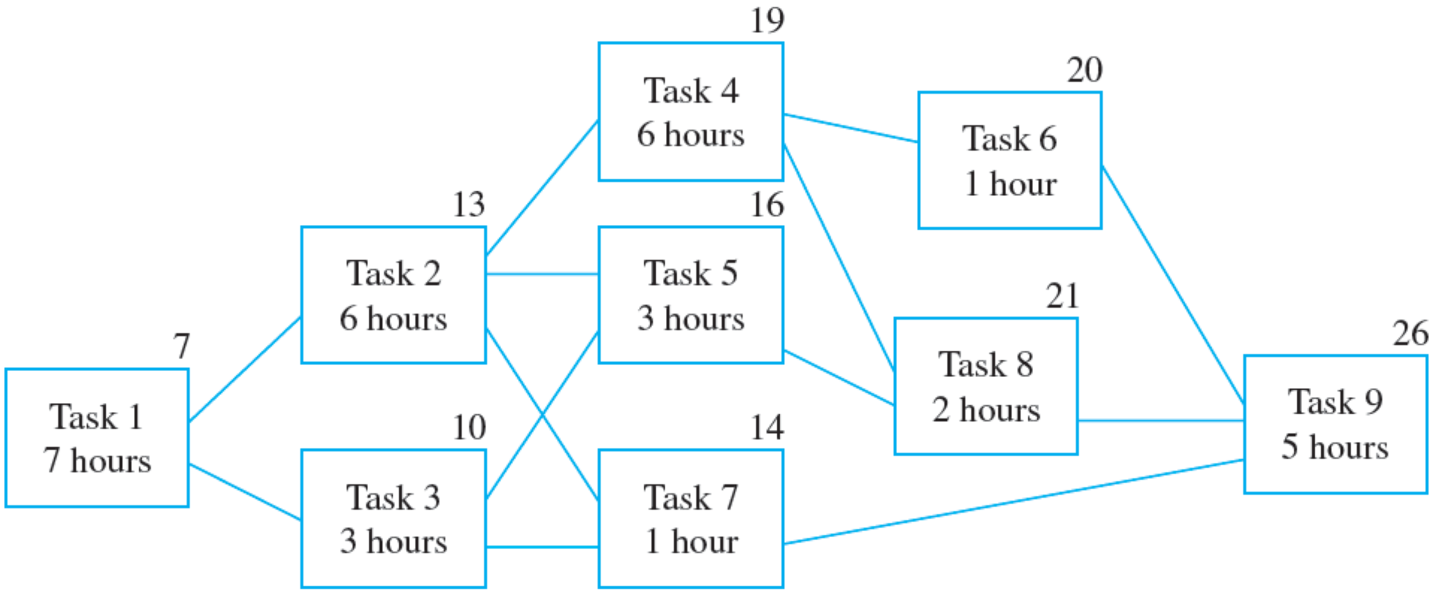
The minimum times to complete tasks 2 and 3, starting from the beginning of the assembly process, are 13 and 10 hours respectively.

Since *both* tasks must be completed before task 5 can be started, the minimum time to complete task 5, starting from the beginning, is the time needed for task 5 itself (3 hours) plus the *maximum* of the times to complete tasks 2 and 3 (13 hours), and this equals 3 + 13 = 16 hours.

Thus, you should place the number 16 above the box for task 5.

Example 8.5.12 – *A Job Scheduling Problem* continued

The same reasoning leads you to place a 14 above the box for task 7. Similarly, you can place a 19 above the box for task 4, a 20 above the box for task 6, a 21 above the box for task 8, and a 26 above the box for task 9, as shown below.



Example 8.5.12 – *A Job Scheduling Problem* continued

This analysis shows that at least 26 hours are required to complete task 9 starting from the beginning of the assembly process.

When task 9 is finished, the assembly is complete, so 26 hours is the minimum time needed to accomplish the whole process.

Note that the minimum time required to complete tasks 1, 2, 4, 8, and 9 in sequence is exactly 26 hours.

Example 8.5.12 – *A Job Scheduling Problem* continued

This means that a delay in performing any one of these tasks causes a delay in the total time required for assembly of the product.

For this reason, the path through tasks 1, 2, 4, 8, and 9 is called a **critical path**.

Thank You!